INCH-POUND

MS51519B
17 March 2016
SUPERSEDING
MS51519A
17 January 1979

DETAIL SPECIFICATION SHEET

NIPPLE, TUBE, LARGE HEX
FOR USE WITH O-RING, 37 DEGREE FLARED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Inactive for new design after 17 August 1999. For new design, use SAE-J514.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.

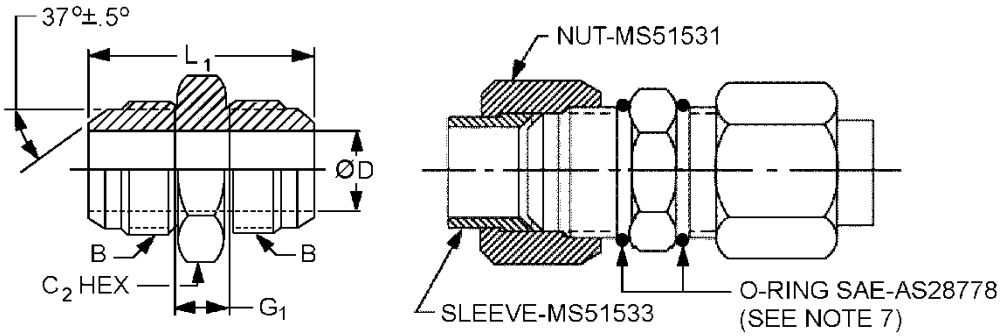


FIGURE 1. Nipple tube.

AMSC N/A

FSC 4730

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

Suffix designator		Tube OD nom.	B Straight thread	C Hex flat Nom.	D diameter	
Assembly	Body (see note 8)				Basic inches (mm)	Tolerance inches (mm)
A2	B2	.1250 (3.175)	.3125-24 UNF-2A	7/16	.062 (1.57)	±.003 (0.08)
A3	B3	.1875 (4.763)	.3750-24 UNF-2A	7/16	.125 (3.18)	
A4	B4	.2500 (6.350)	.4375-20 UNF-2A	1/2	.172 (4.37)	
A5	B5	.3125 (7.936)	.5000-20 UNF-2A	9/16	.234 (5.94)	
A6	B6	.3750 (9.525)	.5625-18 UNF-2A	5/8	.297 (7.54)	±.004 (0.10)
A8	B8	.5000 (12.700)	.7500-16 UNF-2A	13/16	.391 (9.93)	
A10	B10	.6250 (15.875)	.8750-14 UNF-2A	15/16	.484 (12.29)	
A12	B12	.7500 (19.050)	1.0625-12 UN-2A	1 1/8	.609 (15.47)	±.005 (0.13)
A14	B14	.8750 (22.225)	1.1875-12 UN-2A	1 1/4	.718 (18.24)	
A16	B16	1.0000 (25.400)	1.3125-12 UN-2A	1 3/8	.844 (21.44)	±.007 (0.18)
A20	B20	1.2500 (31.750)	1.6250-12 UN-2A	1 11/16	1.078 (27.38)	+.008 -.005 (+0.20 -0.13)
A24	B24	1.5000 (38.100)	1.8750-12 UN-2A	2	1.312 (33.32)	
A32	B32	2.0000 (50.800)	2.500-12 UN-2A	2 5/8	1.781 (45.24)	+.010 -.005 (+0.25 -.013)

Suffix designator		G_1 inches (mm) ±.002 (0.05)	L_1 inches (mm) ±.020 (0.51)
Assembly	Body		
A2	B2	.250 (6.35)	1.170 (29.72)
A3	B3	.313 (7.95)	1.230 (31.24)
A4	B4	.364 (9.25)	1.370 (34.80)
A5	B5	.427 (10.85)	1.370 (34.80)
A6	B6	.482 (12.24)	1.410 (35.81)
A8	B8	.660 (16.76)	1.620 (41.15)
A10	B10	.773 (19.63)	1.880 (47.75)
A12	B12	.945 (24.00)	2.160 (54.86)
A14	B14	1.070 (27.18)	2.210 (56.13)
A16	B16	1.195 (30.35)	2.250 (57.15)
A20	B20	1.507 (38.28)	2.430 (61.72)
A24	B24	1.756 (44.60)	2.750 (69.85)
A32	B32	2.381 (60.48)	3.400 (86.36)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Break all sharp edges and remove all burrs and slivers.
4. Dimensions and tolerances not shown shall be in accordance with SAE-J514 for 37° flared fittings.
5. The drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.
6. Threads shall be in accordance with ASME B1.1.
7. Procure the O-rings separately and install before installation of the fittings.
8. Bodies will be used for production only and shall not be stocked for maintenance.

FIGURE 1. Plug - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Fittings shall be as specified on figure 1 and in tables I and II.

Components of the assemblies shall be of the same material and finish.

Materials shall be in accordance with MIL-DTL-18866 and table I.

TABLE I. Materials.

Material	Form	Specification	Alloy
Carbon steel	Bar	SAE-J403	1117, 1126, 1140, 1213, 1215, or 12L14
	Forgings		
Chrome-molybdenum steel	Bars	SAE-AMS6370	4130
	Forgings	SAE-AMS6382	
		SAE-AMS6370	
Corrosion resistant steel	Bars and forgings	ASTM A276/A246M	304, 304L, 316, or 321
		ASTM A564/A564M	XM-12 (15-5 PH) UNS S15500 or 630 (17-4 PH) UNS S17400
		SAE-AMS5639	UNS S30400
		SAE-AMS5645	UNS S32100
		SAE-AMS5647	UNS S30403
		SAE-AMS5743	UNS S35500
	Bar	ASTM A582/A582M	UNS S30300
Nickel-copper alloy	Bar	ASTM B164 QQ-N-281	UNS N04400
High-chromium nickel alloy	Bar	ASTM B166	UNS N06690
	Forgings	ASTM B564	
Titanium 1/	Bars	SAE-AMS4928	6AI-4V annealed
	Forgings		

1/ Titanium shall not be used in oxygen or potable water systems.

Finish. Finishes shall be as specified in table II. All platings shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of corrosion after 96 hours of salt spray. Fluid passages, other openings and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

TABLE II. Material and finish identification codes.

PIN code material/plating finish	Material	Plating finish
Blank	Steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 2 or SAE-AMS-QQ-P-416, type II, class 2. 1/
H	Steel	Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC.
J	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 3/
M	Nickel-copper alloy UNS N04400	No additional finish.
N	High-chromium nickel alloy UNS N06690	No additional finish.
P	Steel	Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 4. 2/
R	Steel	Zinc plating in accordance with ASTM B633; type VI, Fe/Zn 12. 5/
S	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, method 1, type 6 or 7.
T	Titanium	Anodize in accordance with SAE-AMS2488 type 2. 4/
TF	Titanium	Fluoride phosphate in accordance with SAE-AMS2486. 4/
Z	Steel	Zinc plating in accordance with ASTM B633; type II or III, Fe/Zn 12, or ASTM B695, type II, class 12. 5/
ZC	Steel	Zinc may be any zinc plating from PIN codes H, J, and R with a colored chromate coating 5/

1/ Embrittlement test need not be run. Cadmium shall not be used in oxygen or potable water systems.
2/ Hexavalent chromium free. Finish shall be ROHS compliant.
3/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum coating thickness.
4/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to FED-STD-595 colors 36076 through 36293.
5/ Not for use in aircraft. Requires approval from the Program Officer for all applications.

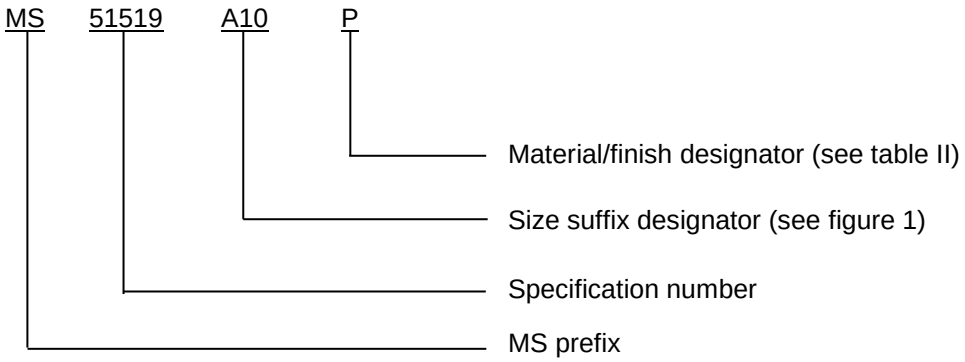
Trivalent wrenchability. When the finish has been damaged due to poor wrenchability, the surface of the connector shall be touched up using one of the brush plating processes below as appropriate to primary finish. The term “trivalent wrenchability” is used to evaluate the ability of the finish to withstand abrasion from an excessive amount of wrenching.

a. Brush plating of hard chromium by electrodeposition shall be in accordance with SAE-AMS-2451/5.
b. Brush plating of medium-hardness, low stress nickel by electrodeposition shall be in accordance with SAE-AMS-2451/9.

Maximum operating pressure. Maximum operating pressure shall be in accordance with SAE-J514.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

PIN: The PIN consists of the letters “MS”, the specification number, a letter and number for nipple size, and a letter for material finish designator.



PIN example: MS51519A10P indicates a nipple, tube, .6250 inch (15.875 mm), steel zinc phosphate finish.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Table III below provides a detailed cross-reference of inactive MS51518 PIN’s and for new design SAE-J514 PIN’s.

MS51518 parts have straight threads in accordance with ASME B1.1 the SAE parts have straight threads in accordance with SAE-J425.

Plating “P” SAE allows a range of nickel from 6% to 20%. Below 12%, ZnNi is not much better than zinc plating, which is less expensive and easier to apply. Above 16%, ZnNi becomes more cathodic and no longer acts as a sacrificial coating - if a high nickel coating is damaged the steel beneath the coating will corrode at an accelerated rate.

The part substitutions specified in table III are intended for most ground vehicle applications and are not intended for use in aerospace applications.

Users are cautioned to evaluate replacement parts for their particular application.

CAUTION: The superseding information is valid as of the date of this specification and may be superseded by subsequent revisions of the superseding document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

TABLE III. MS51519 to SAE-J514 cross reference.

Inactive for new design MS51519- PIN		TUBE O.D.	For new design SAE-J514 PIN	
Assembly	Body		Assembly	Body
MS51519A2	MS51519B2	1/8	Not available	Not available
MS51519A2H	MS51519B2H	1/8	Not available	Not available
MS51519A2J	MS51519B2J	1/8	Not available	Not available
MS51519A2M	MS51519B2M	1/8	J514-2-2-070119MA	J514-2-2-070119MB
MS51519A2N	MS51519B2N	1/8	J514-2-2-070119NA	J514-2-2-070119NB
MS51519A2P	MS51519B2P	1/8	J514-2-2-070119PA	J514-2-2-070119PB
MS51519A2R	MS51519B2R	1/8	Not available	Not available
MS51519A2S	MS51519B2S	1/8	J514-2-2-070119SA	J514-2-2-070119SB
MS51519A2T	MS51519B2T	1/8	J514-2-2-070119TA	J514-2-2-070119TB
MS51519A2TF	MS51519B2TF	1/8	Not available	Not available
MS51519A2Z	MS51519B2Z	1/8	J514-2-2-070119ZA	J514-2-2-070119ZB
MS51519A2ZC	MS51519B2ZC	1/8	Not available	Not available
MS51519A3	MS51519B3	3/16	Not available	Not available
MS51519A3H	MS51519B3H	3/16	Not available	Not available
MS51519A3J	MS51519B3J	3/16	Not available	Not available
MS51519A3M	MS51519B3M	3/16	J514-3-3-070119MA	J514-3-3-070119MB
MS51519A3N	MS51519B3N	3/16	J514-3-3-070119NA	J514-3-3-070119NB
MS51519A3P	MS51519B3P	3/16	J514-3-3-070119PA	J514-3-3-070119PB
MS51519A3R	MS51519B3R	3/16	Not available	Not available
MS51519A3S	MS51519B3S	3/16	J514-3-3-070119SA	J514-3-3-070119SB
MS51519A3T	MS51519B3T	3/16	J514-3-3-070119TA	J514-3-3-070119TB
MS51519A3TF	MS51519B3TF	3/16	Not available	Not available
MS51519A3Z	MS51519B3Z	3/16	J514-3-3-070119ZA	J514-3-3-070119ZB
MS51519A3ZC	MS51519B3ZC	3/16	Not available	Not available
MS51519A4	MS51519B4	1/4	Not available	Not available
MS51519A4H	MS51519B4H	1/4	Not available	Not available
MS51519A4J	MS51519B4J	1/4	Not available	Not available
MS51519A4M	MS51519B4M	1/4	J514-4-4-070119MA	J514-4-4-070119MB
MS51519A4N	MS51519B4N	1/4	J514-4-4-070119NA	J514-4-4-070119NB
MS51519A4P	MS51519B4P	1/4	J514-4-4-070119PA	J514-4-4-070119PB
MS51519A4R	MS51519B4R	1/4	Not available	Not available
MS51519A4S	MS51519B4S	1/4	J514-4-4-070119SA	J514-4-4-070119SB
MS51519A4T	MS51519B4T	1/4	J514-4-4-070119TA	J514-4-4-070119TB
MS51519A4TF	MS51519B4TF	1/4	Not available	Not available
MS51519A4Z	MS51519B4Z	1/4	J514-4-4-070119ZA	J514-4-4-070119ZB
MS51519A4ZC	MS51519B4ZC	1/4	Not available	Not available
MS51519A5	MS51519B5	5/16	Not available	Not available
MS51519A5H	MS51519B5H	5/16	Not available	Not available
MS51519A5J	MS51519B5J	5/16	Not available	Not available
MS51519A5M	MS51519B5M	5/16	J514-5-5-070119MA	J514-5-5-070119MB
MS51519A5N	MS51519B5N	5/16	J514-5-5-070119NA	J514-5-5-070119NB
MS51519A5P	MS51519B5P	5/16	J514-5-5-070119PA	J514-5-5-070119PB
MS51519A5R	MS51519B5R	5/16	Not available	Not available
MS51519A5S	MS51519B5S	5/16	J514-5-5-070119SA	J514-5-5-070119SB
MS51519A5T	MS51519B5T	5/16	J514-5-5-070119TA	J514-5-5-070119TB
MS51519A5TF	MS51519B5TF	5/16	Not available	Not available
MS51519A5Z	MS51519B5Z	5/16	J514-5-5-070119ZA	J514-5-5-070119ZB
MS51519A5ZC	MS51519B5ZC	5/16	Not available	Not available

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

TABLE III. MS51519 to SAE-J514 cross reference. - Continued.

Inactive for new design MS51519- PIN		TUBE O.D.	For new design SAE-J514 PIN 1/	
Assembly	Body		Assembly	Body
MS51519A6	MS51519B6	3/8	Not available	Not available
MS51519A6H	MS51519B6H	3/8	Not available	Not available
MS51519A6J	MS51519B6J	3/8	Not available	Not available
MS51519A6M	MS51519B6M	3/8	J514-6-6-070119MA	J514-6-6-070119MB
MS51519A6N	MS51519B6N	3/8	J514-6-6-070119NA	J514-6-6-070119NB
MS51519A6P	MS51519B6P	3/8	J514-6-6-070119PA	J514-6-6-070119PB
MS51519A6R	MS51519B6R	3/8	Not available	Not available
MS51519A6S	MS51519B6S	3/8	J514-6-6-070119SA	J514-6-6-070119SB
MS51519A6T	MS51519B6T	3/8	J514-6-6-070119TA	J514-6-6-070119TB
MS51519A6TF	MS51519B6TF	3/8	Not available	Not available
MS51519A6Z	MS51519B6Z	3/8	J514-6-6-070119ZA	J514-6-6-070119ZB
MS51519A6ZC	MS51519B6ZC	3/8	Not available	Not available
MS51519A8	MS51519B8	1/2	Not available	Not available
MS51519A8H	MS51519B8H	1/2	Not available	Not available
MS51519A8J	MS51519B8J	1/2	Not available	Not available
MS51519A8M	MS51519B8M	1/2	J514-8-8-070119MA	J514-8-8-070119MB
MS51519A8N	MS51519B8N	1/2	J514-8-8-070119NA	J514-8-8-070119NB
MS51519A8P	MS51519B8P	1/2	J514-8-8-070119PA	J514-8-8-070119PB
MS51519A8R	MS51519B8R	1/2	Not available	Not available
MS51519A8S	MS51519B8S	1/2	J514-8-8-070119SA	J514-8-8-070119SB
MS51519A8T	MS51519B8T	1/2	J514-8-8-070119TA	J514-8-8-070119TB
MS51519A8TF	MS51519B8TF	1/2	Not available	Not available
MS51519A8Z	MS51519B8Z	1/2	J514-8-8-070119ZA	J514-8-8-070119ZB
MS51519A8ZC	MS51519B8ZC	1/2	Not available	Not available
MS51519A10	MS51519B10	5/8	Not available	Not available
MS51519A10H	MS51519B10H	5/8	Not available	Not available
MS51519A10J	MS51519B10J	5/8	Not available	Not available
MS51519A10M	MS51519B10M	5/8	J514-10-10-070119MA	J514-10-10-070119MB
MS51519A10N	MS51519B10N	5/8	J514-10-10-70119NA	J514-10-10-070119NB
MS51519A10P	MS51519B10P	5/8	J514-10-10-70119PA	J514-10-10-070119PB
MS51519A10R	MS51519B10R	5/8	Not available	Not available
MS51519A10S	MS51519B10S	5/8	J514-10-10-070119SA	J514-10-10-070119SB
MS51519A10T	MS51519B10T	5/8	J514-10-10-070119TA	J514-10-10-070119TB
MS51519A10TF	MS51519B10TF	5/8	Not available	Not available
MS51519A10Z	MS51519B10Z	5/8	J514-10-10-70119ZA	J514-10-10-070119ZB
MS51519A10ZC	MS51519B10ZC	5/8	Not available	Not available
MS51519A12	MS51519B12	3/4	Not available	Not available
MS51519A12H	MS51519B12H	3/4	Not available	Not available
MS51519A12J	MS51519B12J	3/4	Not available	Not available
MS51519A12M	MS51519B12M	3/4	J514-12-12-070119MA	J514-12-12-070119MB
MS51519A12N	MS51519B12N	3/4	J514-12-12-070119NA	J514-12-12-070119NB
MS51519A12P	MS51519B12P	3/4	J514-12-12-070119PA	J514-12-12-070119PB
MS51519A12R	MS51519B12R	3/4	Not available	Not available
MS51519A12S	MS51519B12S	3/4	J514-12-12-070119SA	J514-12-12-070119SB
MS51519A12T	MS51519B12T	3/4	J514-12-12-070119TA	J514-12-12-070119TB
MS51519A12TF	MS51519B12TF	3/4	Not available	Not available
MS51519A12Z	MS51519B12Z	3/4	J514-12-12-070119ZA	J514-12-12-070119ZB
MS51519A12ZC	MS51519B12ZC	3/4	Not available	Not available

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

TABLE III. MS51519 to SAE-J514 cross reference. - Continued.

Inactive for new design MS51519- PIN		TUBE O.D.	For new design SAE-J514 PIN 1/	
Assembly	Body		Assembly	Body
MS51519A14	MS51519B14	7/8	Not available	Not available
MS51519A14H	MS51519B14H	7/8	Not available	Not available
MS51519A14J	MS51519B14J	7/8	Not available	Not available
MS51519A14M	MS51519B14M	7/8	J514-14-14-070119MA	J514-14-14-070119MB
MS51519A14N	MS51519B14N	7/8	J514-14-14-070119NA	J514-14-14-070119NB
MS51519A14P	MS51519B14P	7/8	J514-14-14-070119PA	J514-14-14-070119PB
MS51519A14R	MS51519B14R	7/8	Not available	Not available
MS51519A14S	MS51519B14S	7/8	J514-14-14-070119SA	J514-14-14-070119SB
MS51519A14T	MS51519B14T	7/8	J514-14-14-070119TA	J514-14-14-070119TB
MS51519A14TF	MS51519B14TF	7/8	Not available	Not available
MS51519A14Z	MS51519B14Z	7/8	J514-14-14-070119ZA	J514-14-14-070119ZB
MS51519A14ZC	MS51519B14ZC	7/8	Not available	Not available
MS51519A16	MS51519B16	1	Not available	Not available
MS51519A16H	MS51519B16H	1	Not available	Not available
MS51519A16J	MS51519B16J	1	Not available	Not available
MS51519A16M	MS51519B16M	1	J514-16-16-070119MA	J514-16-16-070119MB
MS51519A16N	MS51519B16N	1	J514-16-16-070119NA	J514-16-16-070119NB
MS51519A16P	MS51519B16P	1	J514-16-16-070119PA	J514-16-16-070119PB
MS51519A16R	MS51519B16R	1	Not available	Not available
MS51519A16S	MS51519B16S	1	J514-16-16-070119SA	J514-16-16-070119SB
MS51519A16T	MS51519B16T	1	J514-16-16-070119TA	J514-16-16-070119TB
MS51519A16TF	MS51519B16TF	1	Not available	Not available
MS51519A16Z	MS51519B16Z	1	J514-16-16-070119ZA	J514-16-16-070119ZB
MS51519A16ZC	MS51519B16ZC	1	Not available	Not available
MS51519A20	MS51519B20	1 1/4	Not available	Not available
MS51519A20H	MS51519B20H	1 1/4	Not available	Not available
MS51519A20J	MS51519B20J	1 1/4	Not available	Not available
MS51519A20M	MS51519B20M	1 1/4	J514-20-20-070119MA	J514-20-20-070119MB
MS51519A20N	MS51519B20N	1 1/4	J514-20-20-070119NA	J514-20-20-070119NB
MS51519A20P	MS51519B20P	1 1/4	J514-20-20-070119PA	J514-20-20-070119PB
MS51519A20R	MS51519B20R	1 1/4	Not available	Not available
MS51519A20S	MS51519B20S	1 1/4	J514-20-20-070119SA	J514-20-20-070119SB
MS51519A20T	MS51519B20T	1 1/4	J514-20-20-070119TA	J514-20-20-070119TB
MS51519A20TF	MS51519B20TF	1 1/4	Not available	Not available
MS51519A20Z	MS51519B20Z	1 1/4	J514-20-20-070119ZA	J514-20-20-070119ZB
MS51519A20ZC	MS51519B20ZC	1 1/4	Not available	Not available
MS51519B24	MS51519A24	1 1/2	Not available	Not available
MS51519B24H	MS51519A24H	1 1/2	Not available	Not available
MS51519B24J	MS51519A24J	1 1/2	Not available	Not available
MS51519B24M	MS51519A24M	1 1/2	J514-24-24-070119MA	J514-24-24-070119MB
MS51519B24N	MS51519A24N	1 1/2	J514-24-24-070119NA	J514-24-24-070119NB
MS51519B24P	MS51519A24P	1 1/2	J514-24-24-070119PA	J514-24-24-070119PB
MS51519B24R	MS51519A24R	1 1/2	Not available	Not available
MS51519B24S	MS51519A24S	1 1/2	J514-24-24-070119SA	J514-24-24-070119SB
MS51519B24T	MS51519A24T	1 1/2	J514-24-24-070119TA	J514-24-24-070119TB
MS51519B24TF	MS51519A24TF	1 1/2	Not available	Not available
MS51519B24Z	MS51519A24Z	1 1/2	J514-24-24-070119ZA	J514-24-24-070119ZB
MS51519B24ZC	MS51519A24ZC	1 1/2	Not available	Not available

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.

TABLE III. MS51519 to SAE-J514 cross reference. - Continued.

Inactive for new design MS51519- PIN		TUBE O.D.	For new design SAE-J514 PIN 1/	
Assembly	Body		Assembly	Body
MS51519B32	MS51519A32	2	Not available	Not available
MS51519B32H	MS51519A32H	2	Not available	Not available
MS51519B32J	MS51519A32J	2	Not available	Not available
MS51519B32M	MS51519A32M	2	J514-32-32-070119MA	J514-32-32-070119MB
MS51519B32N	MS51519A32N	2	J514-32-32-070119NA	J514-32-32-070119NB
MS51519B32P	MS51519A32P	2	J514-32-32-070119PA	J514-32-32-070119PB
MS51519B32R	MS51519A32R	2	Not available	Not available
MS51519B32S	MS51519A32S	2	J514-32-32-070119SA	J514-32-32-070119SB
MS51519B32T	MS51519A32T	2	J514-32-32-070119TA	J514-32-32-070119TB
MS51519B32TF	MS51519A32TF	2	Not available	Not available
MS51519B32Z	MS51519A32Z	2	J514-32-32-070119ZA	J514-32-32-070119ZB
MS51519B32ZC	MS51519A32ZC	2	Not available	Not available

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-18866, this document references the following:

FED-STD-595/36076	FED-STD-595/36293	ASTM B633	SAE-AMS5645
FED-STD-595/36081	MIL-DTL-16232	ASTM B695	SAE-AMS5647
FED-STD-595/36099	MS51531	ASTM F1136/F1136M	SAE-AMS5743
FED-STD-595/36118	MS51533	SAE-AMS-C-81562	SAE-AMS6370
FED-STD-595/36134	QQ-N-281	SAE-AMS-QQ-P-416	SAE-AMS6382
FED-STD-595/36152	ASME B1.1	SAE-AMS2417	SAE-AS28778
FED-STD-595/36170	ASTM A276/A276M	SAE-AMS2451/5	SAE-J403
FED-STD-595/36173	ASTM A564/A564M	SAE-AMS2451/9	SAE-J425
FED-STD-595/36176	ASTM A582/A582M	SAE-AMS2486	SAE-J514
FED-STD-595/36231	ASTM B117	SAE-AMS2488	
FED-STD-595/36251	ASTM B164	SAE-AMS2700	
FED-STD-595/36270	ASTM B166	SAE-AMS4928	
FED-STD-595/36280	ASTM B564	SAE-AMS5639	

CONCLUDING MATERIAL

Custodians:
Army - AR
Navy - OS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2016-019)

Review activities:
Army - AT, MI
Navy - CG, MC, SA, SH
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:05Z
Check the source to verify that this is the current version before use.